

15 Instrument-Assisted Soft Tissue Mobilization

Another treatment that is gaining popularity is Instrument Assisted Soft Tissue Mobilization (IASTM). Although recently popular, its origins were many millennia in the past and related to Gua Sha, which was a scraping technique leading to light bruising (40). IASTM is similar to manual massage or foam rolling as it involves placing stress and strain on the tissues. This is accomplished with metal tools that conform to specific body areas and permit individuals to apply pressure. It is suggested that IASTM allows more direct treatment of tissues than activities like stretching. Similar to the rationale presented for early foam rolling articles, IASTM is purported to be a soft tissue treatment technique that can positively reduce scar tissue and myofascial adhesions. However, again similar to the lack of foam rolling proof (9), the evidence for attenuating or removing myofascial restrictions or trigger points with IASTM is lacking. Whereas some of the purported mechanisms are not clear, the IASTM effects on ROM have been established.

Acute Effects on Range of Motion

There are a variety of studies demonstrating improved range of motion (ROM) with a single session (acute) of IASTM. IASTM-induced increases in ROM have been recorded at the neck (cervical) (1, 35), with shoulder internal rotation (2, 31) and horizontal adduction (31), hamstrings (hip flexion) (13, 30, 39, 41), knee (45) and hip (37), as well as ankle dorsiflexion (19, 33, 44). These ROM improvements were exhibited after IASTM treatment durations of as little as two (37) and five minutes (13, 30). Shoulder internal rotation ROM improvements with IASTM endured 45 minutes after treatment, which was significantly longer than with than kinesiology taping (2). One study reported ROM gains were still significant 24 hours after IASTM treatment (37).

A number of these studies showed similar ROM benefits when employing IASTM compared to foam rolling of the cervical area (1), hip, and knee (37, 45), shoulder internal rotation with flossing (2), therapeutic cupping (13), vibration massage, light hand massage (30), muscle energy technique with the hamstrings (39), stretching for hip flexion active ROM (41), and dorsiflexion



Figure 15.1 Instrument Assisted Soft Tissue Mobilization (IASTM) devices



Figure 15.2 Instrument Assisted Soft Tissue Mobilization (IASTM) use on quadriceps



Figure 15.3 Instrument Assisted Soft Tissue Mobilization (IASTM) use on forearms



Figure 15.4 Instrument Assisted Soft Tissue Mobilization (IASTM) use on calf

ROM (44). IASTM was reported to provide superior shoulder internal rotation ROM than kinesiotaping (2), and neck ROM compared to routine physical therapy (35).

Training Effects on Range of Motion

IASTM can also have enduring effects after chronic treatments. Repetitive or training-related ROM gains have been reported when testing dorsiflexion after six treatments (10), low back after eight treatments (16), shoulder flexion after once a week treatments for three weeks, shoulder horizontal adduction and internal rotation with five minutes per session, two days a week, over a period of four weeks (20), and one IASTM treatment per week for six weeks significantly improved hip adduction ROM. Again, when comparing ROM benefits to other treatments, IASTM outperformed foam rolling and elastic taping for shoulder ROM when implemented once a week for three weeks (36), and was more beneficial than foam rolling or stretching for hip ROM (once per week for six weeks) (46). IASTM provided similar benefits as a conventional physical therapy programme or kinesiotaping in patients with chronic mechanical low back pain over eight sessions (16), and similar results when compared to a combination of IASTM and post-isometric stretch over four weeks (two days per week) for shoulder adduction (20). Another study reported that five sessions of IASTM with three days of recovery between sessions provided greater active triceps surae ROM than percussive massage (with a Theragun) immediately after training as well as five days later (48).

Global effects of IASTM

We have shown that responses to exercise, fatigue, stretching, or foam rolling are not isolated to the affected muscle or limb, but can have global implications for the performance (i.e., muscle strength, power, and endurance) and ROM of other non-exercised or treated muscles or joints (3, 4, 6, 18, 27, 28). Based on this body of evidence one might expect similar global responses with IASTM. Only one study has investigated non-local IASTM responses. Following four weeks of IASTM on the back (trunk), whether it was applied to the upper or lower part of the superficial back line, there was an increase in supine leg raise (hamstrings) ROM (14). This non-local response may be attributed to cortical responses related to central pain-modulatory systems (14) such as the diffuse noxious inhibitory control system (global release of endorphins and enkephalins in response to pain) (32). Furthermore, there are reports of general body relaxation and decreases in myofascial tone with IASTM (14). In addition, the mechanical force transmission through myofascial chains may transfer strain to neighbouring skeletal muscles and fascia (11).

IASTM Effects on Performance

While stretching research has repeatedly highlighted performance impairments when an acute bout of static stretching is prolonged (i.e., >60-s per muscle group) and not conducted in conjunction with dynamic activities (7, 8, 12, 21), foam rolling typically does not hinder subsequent performance (5, 23, 29). As IASTM is more similar to foam rolling than stretching, what effect would a session of IASTM have on subsequent performance? Similar to many foam rolling studies, there were no immediate deficits apparent in vertical jump height, peak power, and velocity (34), Y-balance test (33) or sprinting speed (47) after an IASTM treatment. In contrast, IASTM alone (once a week for three weeks) (36) or an acute bout in combination with flossing (2) augmented throwing performance. Neck function has been reported to be improved with IASTM and foam rolling (three times in one week) (1) as well as following eight treatments that combined IASTM and exercises for the cervical and thoracic area (38). Similarly, eight IASTM treatments significantly improved the functional disability index (16). Hamstrings power was significantly improved after a single session of IASTM but not with stretching (41). Acute rehabilitation exercises combined with IASTM decreased fatigue and improved isokinetic peak torques ($30\text{--}180^{\circ}\cdot\text{s}^{-1}$) (22). When compared to Theragun percussive massage treatment (five sessions with three days recovery between treatments), IASTM elicited greater improvements in the overhead deep squat immediately after treatment as well as five days later (48). In summary, there are no reports yet of subsequent performance decrements after IASTM, with the possibility of performance enhancements when IASTM is conducted alone or in combination with other treatments such as foam rolling or flossing.

Mechanisms

An increase in pain tolerance is often listed as a major underlying mechanism for the increase in ROM with stretching (3, 7, 8, 12) and foam rolling (24, 25, 26, 27). As you might expect, the discomfort of using a metal tool to repeatedly place stress and strain on your muscles and fascia can affect your pain tolerance. In many instances, there are reports of significant pain reduction or increased pain tolerance with a treatment of IASTM for example, in a single session for 45 seconds at a frequency of 60 beats/minute (chronic neck pain) (15), treatments for low back pain over eight sessions (16), with six treatments over three weeks for myofascial trigger points (17), and four weeks of IASTM reducing headache symptoms and cervical alignment in patients with tension type headaches (43). Modulation of pain tolerance is also seen when IASTM is combined with other modalities such as kinesiotape (16), or foam rolling (three times over one week) (1). Four weeks of IASTM combined with stretching was more effective for neck pain management compared to physical therapy (35). When applied individually, IASTM and foam rolling provided three times over one week were both similarly effective for pain tolerance and sensitivity (1). The concept of myofascial chains or meridians is often explained as a mechanism that transfers

mechanical forces, and strains to other regions contributing to decreases in pain and improvements in ROM (11). One study reported that, as IASTM did not alter peak passive torque and muscle stiffness, joint ROM can be improved without altering tissue mechanical properties (19). Thus, similar to stretching and foam rolling, an increase in pain tolerance permits the individual to extend muscles farther with less discomfort in an attempt to increase flexibility.

IASTM may induce inflammation in response to the micro-trauma stress caused by the pressure applied with the metal instruments (39). It has been suggested, but not consistently reported (contradictory results) that the IASTM-induced inflammation would produce accelerated tissue repair, reduce adhesions, induce more rapid collagen formation, and therefore re-establish tissue flexibility (39). In contrast, it has also been proposed that IASTM would decrease inflammation, owing to enhanced lymphatic drainage of toxins (49). However, there is little documentation to support these claims. However, reported increases in blood flow to the treated area (42, 49) would be expected to contribute to an accelerated recovery.

Summary

IASTM provides a similar stimulus to muscle and fascia as foam rolling and thus to be expected provides similar outcomes. IASTM can improve ROM following an acute (minimal volume during a single session of 2–5 minutes) bout as well as demonstrating increased ROM with training (chronic effects occurred with 1–2 sessions per week over 3–6 weeks). This flexibility enhancement is primarily attributed to an increase in pain tolerance as also suggested for stretching and foam rolling. These improvements in ROM are not typically accompanied by performance impairments but typically either no decrements or in some studies performance improvements. IASTM is thus recommended as an adjunct or alternative to other flexibility enhancing techniques.

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